

BMW GLOBAL SALES

Exploratory Data Analysis Report

Dataset coverage: 2010 – 2024
50,000 sales records · 11 models · 6 global regions

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1. Executive Summary

This report presents an exploratory data analysis (EDA) of a BMW global sales dataset covering the period 2010–2024. The dataset contains 50,000 individual sales records across 11 BMW models, 6 global regions, 6 exterior colours, 4 fuel types, and 2 transmission types, together with numeric fields for engine size, mileage, price, and sales volume, plus a High/Low sales classification label.

The analysis combines descriptive statistics, distribution analysis, and visual exploration (time trends, regional and model comparisons, correlation analysis, and classification breakdowns) to characterise the structure of the dataset and surface any patterns relevant to sales performance. The central finding is that sales volume, price, mileage, and engine size are almost entirely uncorrelated with one another, and totals are near-identical across every category — a pattern more consistent with randomly generated data than real historical sales records. This is discussed in detail in Section 7.

2. Objectives

- Assess data quality: completeness, consistency, and structure of the dataset.
- Quantify overall sales trends across the 2010–2024 period.
- Compare performance across regions, models, fuel types, transmissions, and colours.
- Examine relationships between price, mileage, engine size, and sales volume.
- Characterise the High vs. Low sales classification and its drivers.
- Summarise key findings and provide recommendations for further analysis.

3. Dataset Description

The table below summarises the structure of the raw dataset.

Attribute	Value
Rows (records)	50,000
Columns	11
Time period	2010 – 2024 (15 years)
Missing values	0 (fully complete)
Duplicate rows	0
Models	11 (3 Series, 5 Series, 7 Series, X1, X3, X5, X6, M3, M5, i3, i8)

Attribute	Value
Regions	6 (Asia, Europe, North America, Middle East, Africa, South America)
Fuel types	4 (Petrol, Diesel, Hybrid, Electric)
Transmissions	2 (Manual, Automatic)
Colours	6 (Red, Blue, Black, Silver, White, Grey)

Numeric field ranges:

Field	Range
Engine_Size_L	1.0 – 5.0 L
Mileage_KM	0 – 199,996 km
Price_USD	\$20,000 – \$119,998
Sales_Volume	100 – 9,999 units

4. Methodology

The analysis was performed in Python using pandas for data manipulation, and matplotlib/seaborn for visualization, run in a Jupyter/Google Colab environment. The workflow followed these steps:

- Data loading and integrity checks (nulls, duplicates, data types).
- Descriptive statistics for all numeric and categorical fields.
- Aggregation of sales volume by year, region, model, fuel type, transmission, and colour.
- Correlation analysis across engine size, mileage, price, and sales volume.
- Cross-tabulation of the High/Low sales classification against model and price.
- Visualization of all the above using line, bar, pie, box, and heatmap charts.

5. Exploratory Data Analysis — Findings

5.1 Sales Trend Over Time

Total annual sales volume fluctuates between roughly 16.3 million and 17.9 million units per year across 2010–2024, with no sustained upward or downward trend — the year-to-year variation looks like statistical noise around a flat average rather than genuine market growth or decline.

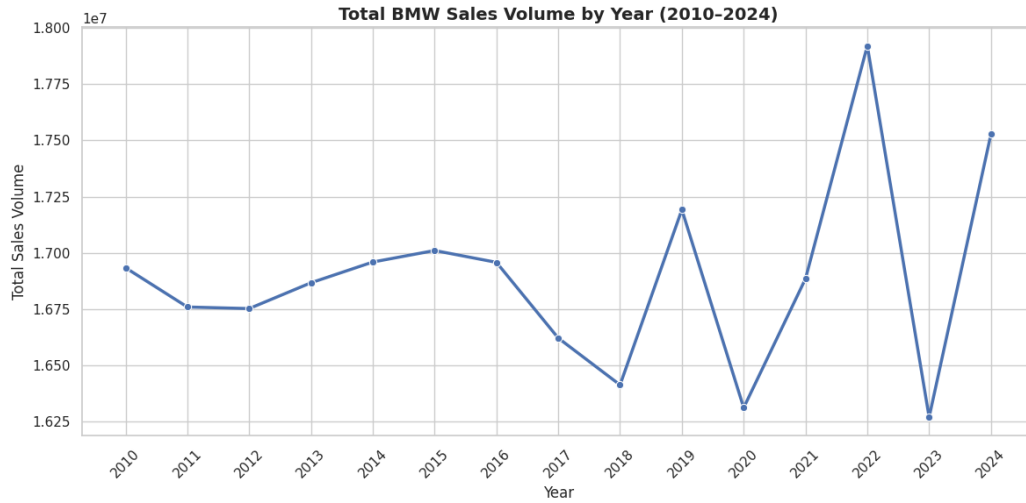


Figure 1. Total BMW sales volume by year, 2010–2024.

5.2 Regional Performance

Total sales volume is almost evenly distributed across all six regions, each contributing between 41.5 and 43.0 million units — a spread of under 3.5% between the highest region (Asia) and the lowest (South America).

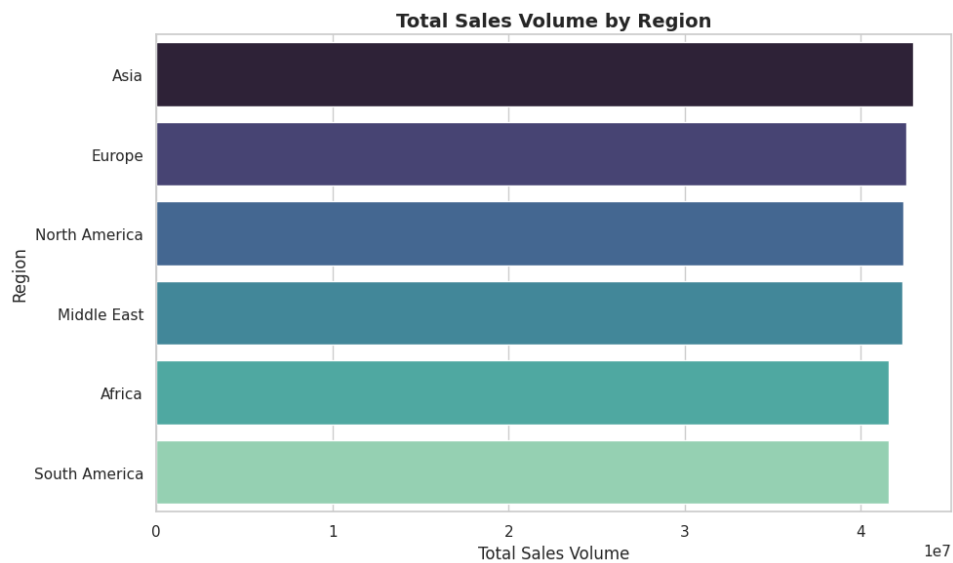


Figure 2. Total sales volume by region.

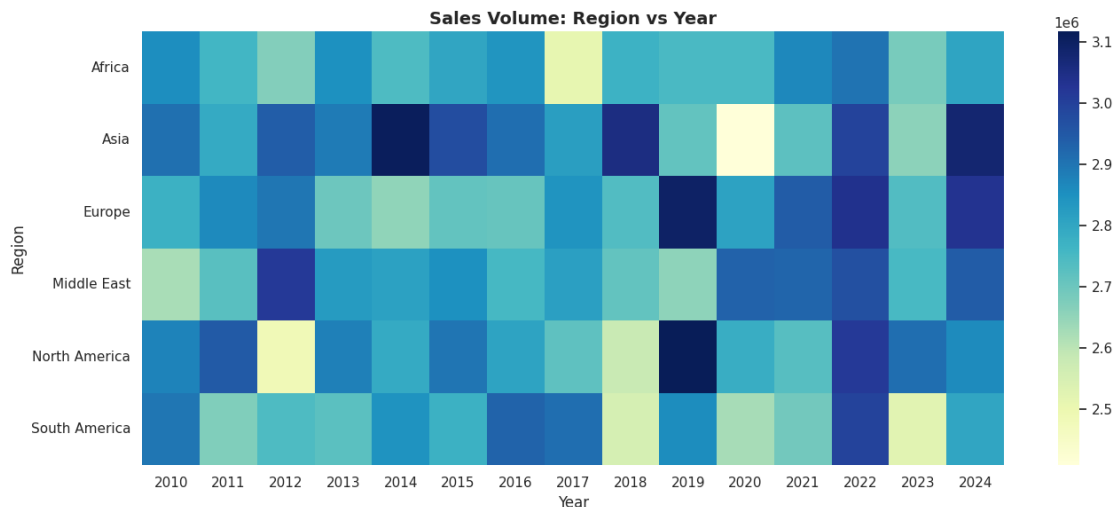


Figure 3. Sales volume by region and year.

5.3 Model-Level Performance

All 11 models post very similar total sales volumes (22.3–23.8 million units each) and near-identical average prices, clustering tightly around \$74,000–\$76,000 regardless of model segment (from the 3 Series up to the 7 Series and M/i performance and EV sub-brands).

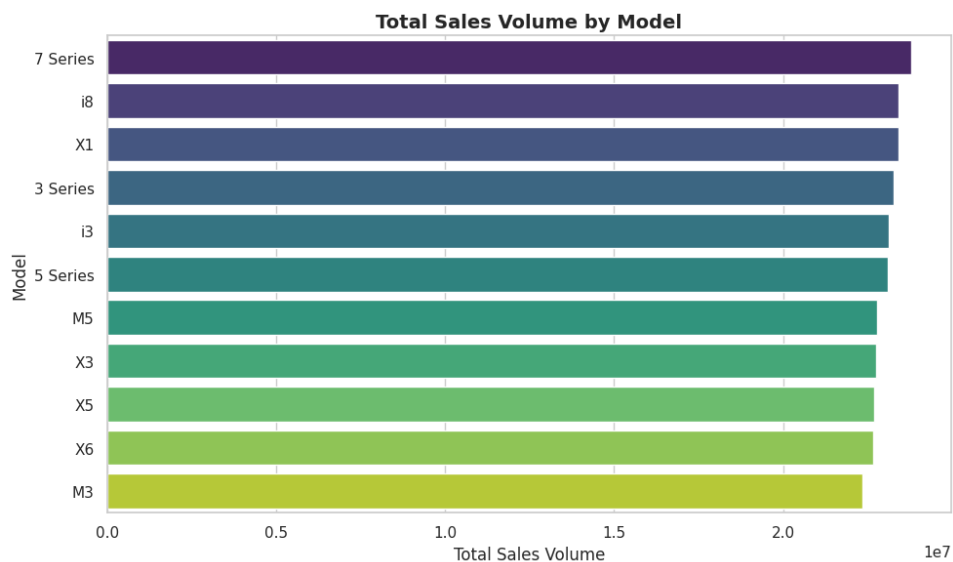


Figure 4. Total sales volume by model.

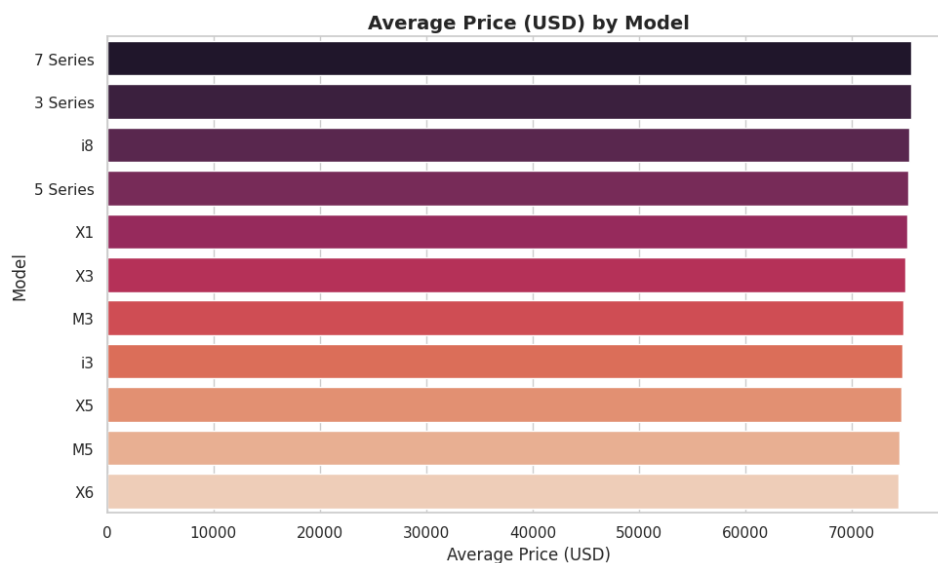


Figure 5. Average price (USD) by model.

5.4 Fuel Type & Transmission

Hybrid, Petrol, Electric, and Diesel variants each account for close to a quarter of total sales volume, and Manual vs. Automatic transmissions split almost exactly 50/50.

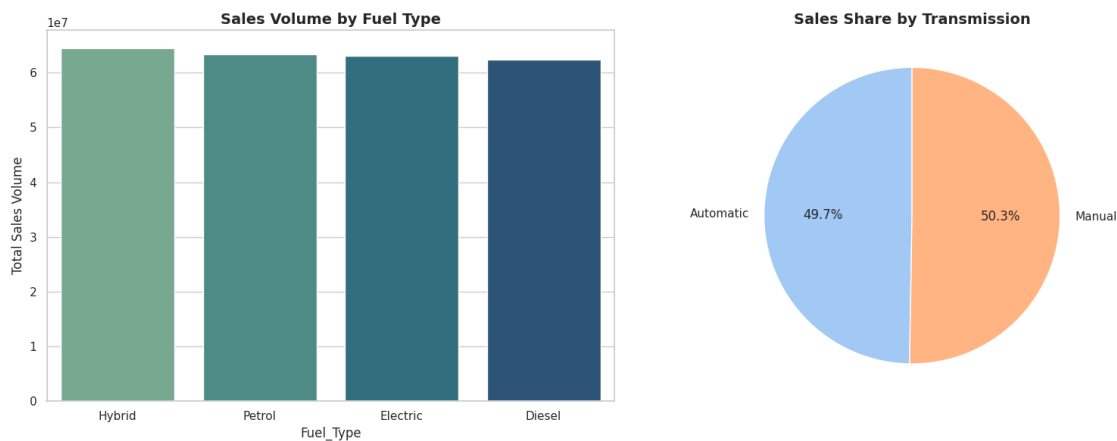


Figure 6. Sales volume by fuel type (left) and transmission share (right).

5.5 Price, Mileage & Engine Size Distributions

Price, mileage, and engine size are all approximately uniformly distributed across their respective ranges, rather than following the bell-shaped or skewed distributions typically seen in real-world pricing or usage data.

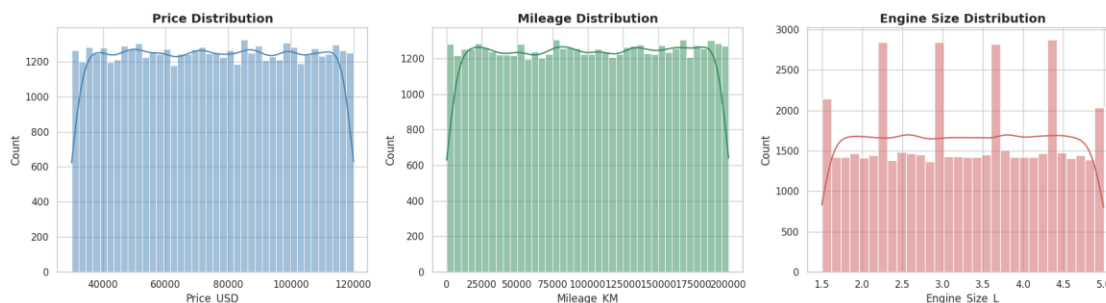


Figure 7. Distributions of price, mileage, and engine size.

5.6 Correlation Analysis

The correlation matrix below shows that engine size, mileage, price, and sales volume are essentially uncorrelated with one another — every pairwise correlation coefficient has an absolute value below 0.005.

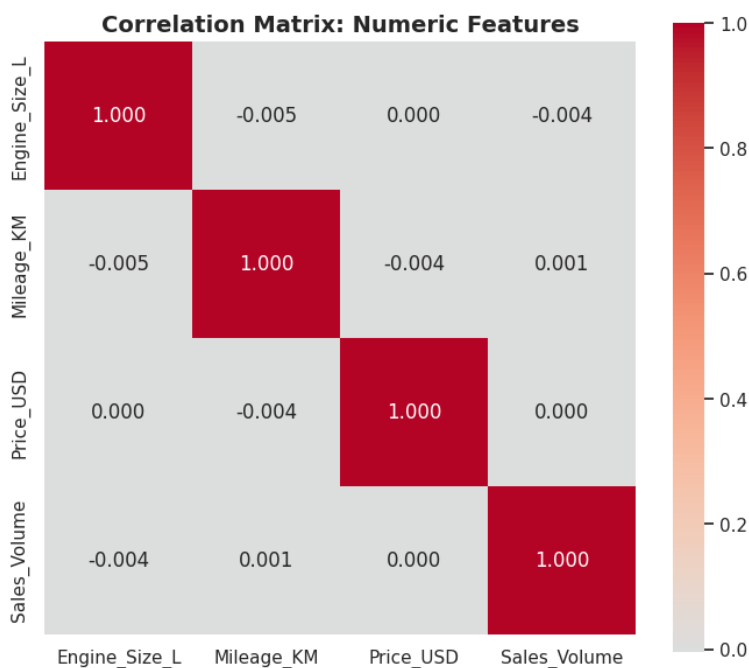


Figure 8. Correlation matrix of numeric features.

5.7 Sales Classification (High vs. Low)

Roughly 30% of records are labelled “High” and 70% “Low”. This split is broadly consistent across models, and price shows very little separation between the High and Low groups — the median and interquartile range of Price_USD are nearly identical for both classes.

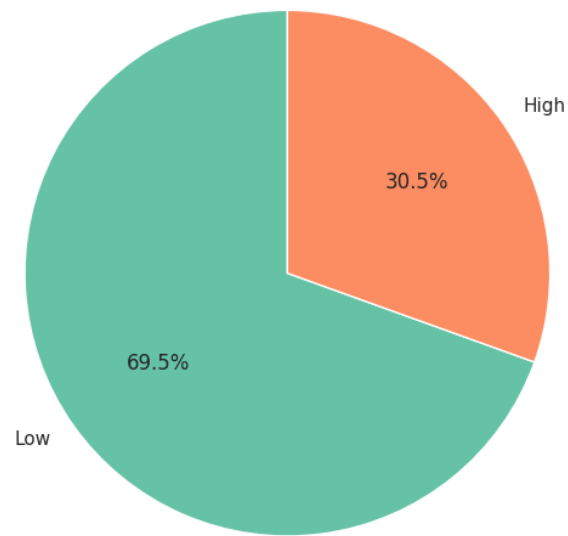
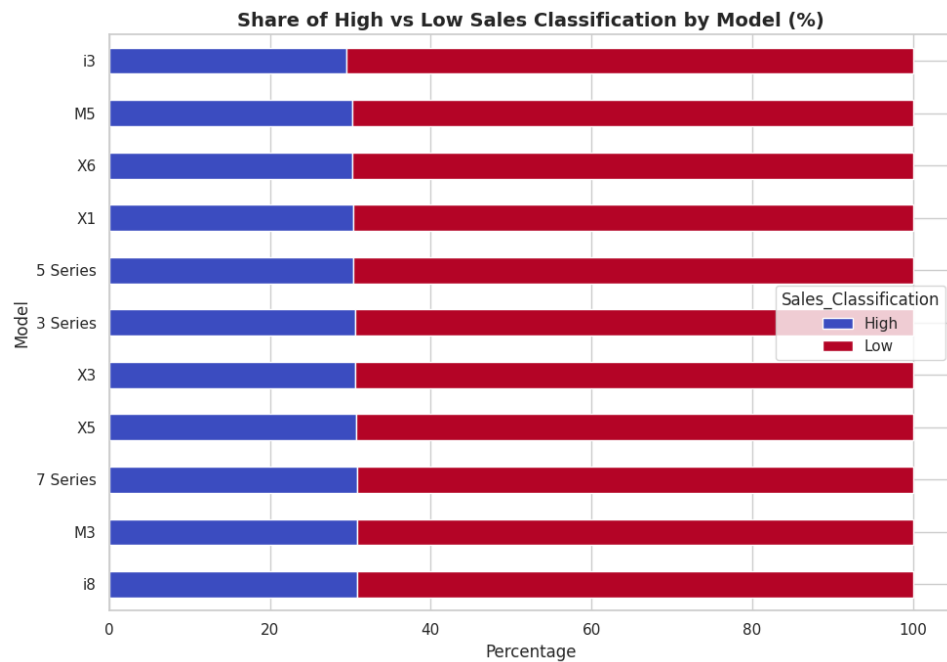
High vs Low Sales Classification Split*Figure 9. High vs. Low sales classification split.**Figure 10. Share of High vs. Low classification by model.*



Figure 11. Price distribution: High vs. Low sales classification.

5.8 Colour Preferences

Sales volume by exterior colour is also evenly spread across all six colours, with no colour standing out as a clear customer favourite.

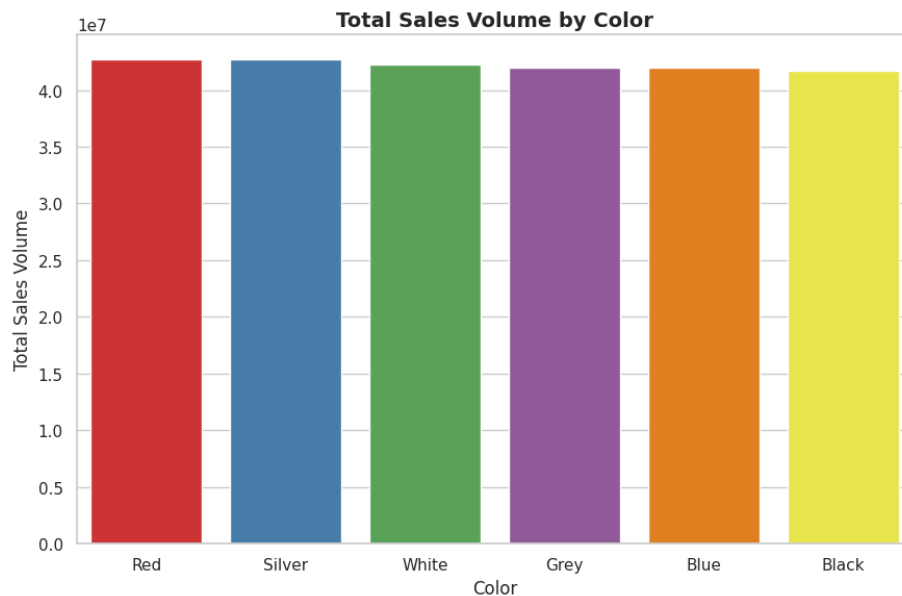


Figure 12. Total sales volume by colour.

5.9 Region × Model Deep Dive

Cross-tabulating region against model confirms the same pattern seen elsewhere: no region-model combination stands out, with cell values distributed evenly across the grid.

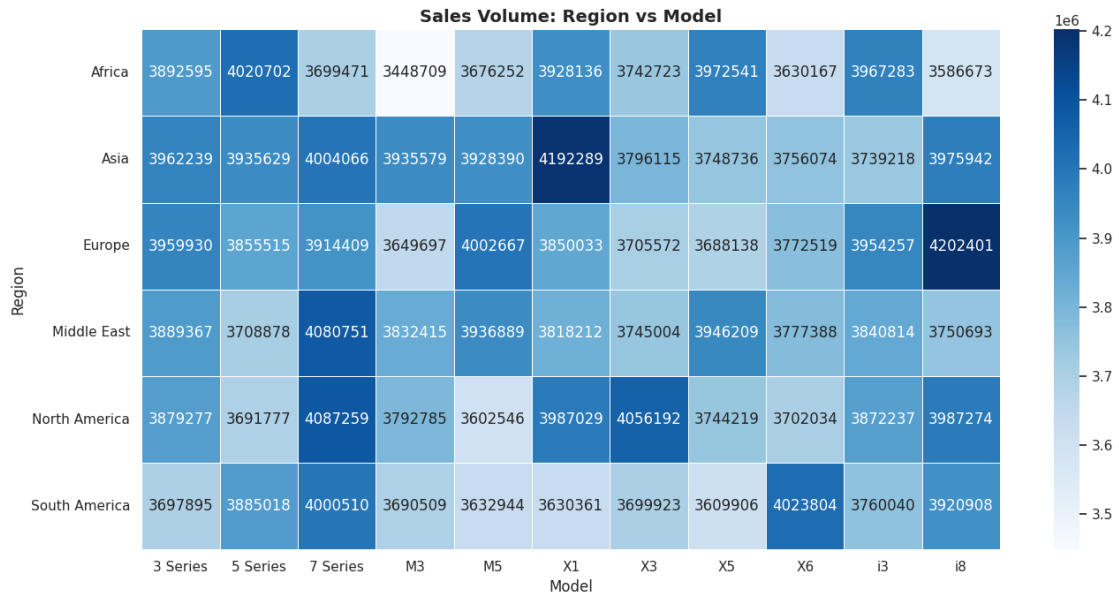


Figure 13. Sales volume by region and model.

6. Summary of Key Statistics

Metric	Value
Total sales volume (all years)	253,375,653 units
Best-performing region	Asia (42.97M units, +3.4% vs. lowest)
Best-performing model	7 Series (23.79M units)
Average price (all models)	~\$75,000
Max correlation with Sales_Volume	$ r = 0.004$ (Mileage_KM)
High-classification share	30.5% of records

7. Discussion — Data Authenticity Note

A consistent pattern emerges throughout this analysis: totals and averages are nearly identical across every categorical breakdown (region, model, fuel type, transmission, colour), and all numeric fields show near-zero correlation with sales volume. In real-world automotive sales data, we would typically expect to see:

- Clear regional leaders driven by market size and demand (e.g. higher volumes in larger markets).
- Premium models (7 Series, M5) commanding higher average prices than entry models (1/3 Series).
- Some relationship between price and sales volume (e.g. price elasticity effects).

- A genuine year-over-year growth or decline trend, reflecting market cycles, launches, or economic conditions.

None of these patterns are present in this dataset. The uniformity of totals and the absence of correlation between price/mileage/engine size and sales volume are strong indicators that this dataset was synthetically or randomly generated rather than sourced from real BMW sales records. This does not diminish its usefulness as a practice dataset for EDA and visualization workflows, but any business conclusions drawn from it should be treated as illustrative only, not reflective of actual BMW market performance.

8. Recommendations

- Treat this dataset as a training/demo dataset for analytics workflows rather than a source of real market insight.
- If real sales insight is required, source data from BMW investor relations reports, national automotive registries, or a verified market research provider.
- If further statistical confirmation is desired, run formal hypothesis tests (ANOVA across regions/models, chi-square on classification vs. categorical fields) to quantitatively confirm the absence of significant relationships noted above.
- For a real dataset, extend this EDA with time-series decomposition, price-elasticity modelling, and a predictive model for the sales classification label.

9. Conclusion

This EDA provided a structured, visual walkthrough of the BMW sales dataset spanning 2010–2024. The dataset is clean and complete, with no missing values or duplicates, and lends itself well to a full range of exploratory techniques. However, the near-uniform distribution of sales across all categories and the absence of meaningful correlations between key variables indicate the data is very likely synthetic. The accompanying Jupyter/Colab notebook (BMW_Sales_EDA.ipynb) reproduces every chart and statistic in this report and can be re-run or extended for further analysis.